CSE STUDY HUB

Emerging Technology

UNIT 4 · Semiconductor & Advanced Computing

What this unit is about

This unit builds understanding of **Semiconductor & Advanced Computing**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Semiconductor materials and properties; conductors vs insulators

2. PN junction and diode/transistor as core electronic components

3. Integrated circuits, logic gates and chip design/fabrication overview

4. Fabrication stages: wafer, lithography, etching, doping and packaging

5. CPU, GPU, parallel and high-performance computing architectures

6. Quantum computing: qubits, superposition and entanglement (overview)

7. Applications, scaling constraints, power, cost and reliability

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 4 NOTES · PART 1

Semiconductor materials and properties; conductors vs insulators

A semiconductor has conductivity between a conductor and insulator and can be controlled by doping. PN junctions form the basis of diodes; transistors act as switches/amplifiers in integrated circuits.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

PN junction and diode/transistor as core electronic components

PN junction and diode/transistor as core electronic components is a syllabus concept in Emerging Technology. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Integrated circuits, logic gates and chip design/fabrication overview

Start with a precise definition of Integrated circuits, logic gates and chip design/fabrication overview. Explain its purpose, the problem it solves, its main components and where it is used. In an exam answer, add one practical example and distinguish it from the nearest related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Semiconductor & Advanced Computing, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 4 NOTES · PART 2

Fabrication stages: wafer, lithography, etching, doping and packaging

A graph contains vertices and edges. An adjacency matrix uses $O(V^2)$ space; an adjacency list uses $O(V+E)$. BFS explores level by level with a queue; DFS explores deeply using recursion or a stack.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

CPU, GPU, parallel and high-performance computing architectures

Treat CPU, GPU, parallel and high-performance computing architectures as a structured model. Name its components in order, explain the responsibility of each component, and show how information moves between them. Include a labelled block diagram and one use case.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Semiconductor & Advanced Computing, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 4 NOTES · PART 3

Quantum computing: qubits, superposition and entanglement (overview)

The OSI model has Physical, Data Link, Network, Transport, Session, Presentation and Application layers. Each layer provides services to the one above and uses the one below; encapsulation adds headers as data moves down.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Applications, scaling constraints, power, cost and reliability

For Applications, scaling constraints, power, cost and reliability, connect the concept to a practical scenario. Identify the input or need, the technology/process used, the output or benefit, and one limitation. Strong examples explain not only what is used but why it is suitable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Semiconductor & Advanced Computing, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Semiconductor materials and properties; conductors vs insulators** — know the definition, core working and one example.

• **PN junction and diode/transistor as core electronic components** — know the definition, core working and one example.

• **Integrated circuits, logic gates and chip design/fabrication overview** — know the definition, core working and one example.

• **Fabrication stages: wafer, lithography, etching, doping and packaging** — know the definition, core working and one example.

• **CPU, GPU, parallel and high-performance computing architectures** — know the definition, core working and one example.

• **Quantum computing: qubits, superposition and entanglement (overview)** — know the definition, core working and one example.

• **Applications, scaling constraints, power, cost and reliability** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Semiconductor & Advanced Computing. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Semiconductor materials and properties; conductors vs insulators and explain its importance.

Q2. Explain the working/steps involved in PN junction and diode/transistor as core electronic components.

Q3. Differentiate between two related concepts from Semiconductor & Advanced Computing.

Q4. Give one practical application of Applications, scaling constraints, power, cost and reliability and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Semiconductor & Advanced Computing.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.